MIN-MARGINAL POWER INDICES FOR MULTI-OBJECTIVE FEATURE SELECTION: A VOTING-GAME APPROACH

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ABSTRACT

Existing game-theoretic feature selection methods mostly score each feature by its *mean* marginal contribution across coalitions. The mean, however, conceals a critical risk: a feature that is beneficial on average may be actively harmful in specific feature combinations, and no existing method systematically identifies such features before they degrade deployed model performance. We address this gap by introducing Min-Marginal Power Indices (MinPPI). Rather than averaging, MinPPI takes the *minimum* marginal contribution of each feature under a binary utility-threshold voting game $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$, where U is any higher-is-better utility (AUC, F_1, R^2, ...). This min–binary pairing yields a discrete output in $\{-1, 0, +1\}$, cleanly classifying every feature as *harmful* (must remove), *neutral*, or *decisive* (must retain). The two formulations coincide and the resulting classification is parameter-free (Properties 1–3); less obviously, a feature pivotal in every winning coalition need not be decisive, so worst-case and average marginals genuinely diverge. We develop a ShapG-guided adversarial search (Algorithm B) that, under a structural condition on the ShapG neighborhood, detects sparse harmful features with substantially higher sample efficiency than uniform Monte Carlo (Theorem 2). Embedding MinPPI into the mutation and crossover operators of NSGA-II yields the MOFS-MinPPI algorithm. Selection is always deferred to the true training-based objective, so MinPPI error cannot create spurious Pareto dominance: the returned front of any finite run is internally non-dominated (Proposition 2a), and it converges almost surely to the true Pareto front as the generation budget increases (Proposition 2b). On five tabular datasets (29–4,225 features), MOFS-MinPPI is competitive with seven baselines—SHAP, ShapG, MinShap, Banzhaf, Powershap, MOFS-SHAP, and NSGA2-uniform—on top-k AUC, while its Pareto-front hypervolume and compression rate at $1\%/3\%/5\%$ AUC-loss thresholds match or exceed the strongest NSGA-II baselines. Ablation confirms that the min operator yields better top-k predictive performance than the mean, that binary v_α produces a principled three-way classification absent from continuous $v(\cdot)$, and that MinPPI guidance ($\beta > 0$) improves Pareto-front quality over unguided search. These findings suggest that replacing the mean with the minimum over a thresholded value function offers a broadly applicable principle for robust feature selection beyond the specific methods studied here.

1 INTRODUCTION

Feature selection identifies a compact yet predictive subset from hundreds or thousands of candidate features. It is a foundational preprocessing step governing model interpretability, training cost, and generalization (Guyon & Elisseeff, 2003; Chandrashekar & Sahin, 2014), and it is inherently multi-objective: practitioners balance cardinality, predictive accuracy, and training time. Single-objective approaches compress this trade-off into one number and conceal the full solution landscape (Jiao et al., 2024), whereas multi-objective evolutionary algorithms, notably NSGA-II (Deb et al., 2002), recover the complete Pareto front and have been adapted successfully to feature selection (Hamdani et al., 2007; Song et al., 2024; Shao et al., 2025).

Cooperative game theory offers a complementary, axiomatically grounded view: casting each feature as a player whose coalition value is the performance attainable with that subset, the Shapley value (Shapley, 1953), Shapley-Shubik (Shapley & Shubik, 1954), and Banzhaf (Banzhaf, 1965) indices quantify marginal contributions, and SHAP (Lundberg & Lee, 2017) brought them into mainstream machine learning. Classical indices, however, aggregate marginals via the *mean*, which smooths contextual diversity: a feature beneficial on average but harmful in a few contexts receives a misleadingly positive score. Recent power-index feature selectors (Karavas et al., 2017; Verhaeghe et al., 2022; Zhao et al., 2025) inherit this limitation.

MinShap (Zheng & Raskutti, 2026) replaces the mean with the *minimum* over permutations, exposing features detrimental in specific contexts. We take this as inspiration but develop our theory independently. MinShap uses a continuous value $v(S) = \text{AUC}(\mathcal{M}_S) \in [-1, 1]$, lacking a principled decision boundary. We instead pair the min with a *binary* utility-threshold formulation (Owen, 1972): $v_\alpha(S) = \mathbb{1}[U(\mathcal{M}_S) \geq \alpha \cdot U_{\text{full}}]$, where U is any task-dependent utility (AUC, accuracy, F_1, R^2). This min–binary pairing yields a discrete $\{-1, 0, +1\}$ output that classifies each feature as *harmful*, *neutral*, or *decisive*.

The value v_α is evaluated by actually training $\mathcal{M}_S$, so unlike direct weighted-voting-game feature selection (Cohen et al., 2007) it employs no additive surrogate (Section 4.5); the MinPPI scores act only as soft guidance for NSGA-II, with Pareto-optimality decided by true model fitness. Synthesizing power indices, the min-marginal operator, and multi-objective search, our contributions are:

1. **Min-Marginal Power Indices (MinPPI).** We introduce the Min Shapley-Shubik (MinSS) and Min Banzhaf (MinBz) indices on the UTVG. Pairing the min operator with the binary v_α makes their basic properties immediate by construction rather than a contribution in themselves: the two indices coincide and take values in $\{-1, 0, +1\}$, giving a *parameter-free* three-way classification into decisive $(+1)$, harmful (-1), and neutral (0) (Properties 1–3). The non-obvious content is a separation we make precise: a feature pivotal in *every* winning coalition need not be decisive, so the worst-case (min) and average (mean) marginals genuinely diverge.
2. **ShapG-Guided Adversarial Search (Algorithm B).** Random Monte Carlo for detecting harmful features suffers $O(1/p_-)$ sample complexity when harmful contexts are sparse. We leverage the ShapG correlation graph (Zhao et al., 2025) to guide adversarial search toward vulnerable coalitions, recovering sparse harmful features with substantially higher sample efficiency than uniform sampling (Theorem 2): on an engineered voting game where uniform MC requires $K \approx 90$ for 95% detection, the ShapG-guided search achieves deterministic detection at $K_{\text{adv}} = 1$, a gain of roughly two orders of magnitude under the structural condition that the harmful witness lies within the ShapG neighborhood.
3. **MOFS-MinPPI Algorithm.** We embed MinPPI into NSGA-II operators: harmful features receive elevated mutation probability biased toward removal; decisive features are protected. Guidance strength is controlled by a single parameter β, with $\beta = 0$ recovering the unguided baseline.
4. **Comprehensive Empirical Validation.** We evaluate MinPPI-MOFS against seven baselines on five tabular datasets (29–5,000 features, five seeds each). Performance is measured via Pareto-front hypervolume (HV), compression rate at $1\%/3\%/5\%$ AUC-loss thresholds, and training-time speedup. Ablation isolates the min vs. mean operator, binary vs. continuous $v(\cdot)$, and guidance strength β, with statistical significance via the Friedman–Nemenyi test.

2 Related Work

2.1 Game-Theoretic Feature Importance

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, dummy attribution; the Shapley-Shubik index (Shapley & Shubik, 1954) specializes it to simple voting games and the Banzhaf index (Banzhaf, 1965) uses uniform coalition weights, both unified probabilistically by Straffin (1977) (see Chalkiadakis et al. (2011) for computation). SHAP (Lundberg & Lee, 2017)

brought Shapley values to mainstream model explanation, and Cohen et al. (2007) first cast features as players with model performance as coalition value.

Interaction and power indices. Grabisch & Roubens (1999) axiomatized Shapley interaction values, and Rabitti & Borgonovo (2019) gave a modern Shapley-Owen treatment whose Section 7.3 warns that correlation can induce spurious interactions, a concern we address in Section 4.5; Tsai et al. (2023) and Muschalik et al. (2024) refine faithfulness and estimation. For feature selection, Karavas et al. (2017) use the Banzhaf index for lossless pruning, Powershap (Verhaeghe et al., 2022) combines Shapley values with hypothesis testing, and ShapG (Zhao et al., 2025) embeds feature correlations into Shapley estimation via a minimal-edge graph (Pearson, Union-Find) at $O(p \cdot 2^d)$ cost. ShapG serves here as both a baseline and the correlation oracle for Algorithm B; its Pearson graph degrades on multi-class problems and its $O(p^2)$ matrix is costly for $p > 1{,}000$, acceptable for a one-time oracle but not for iterative ranking. Song et al. (2016) developed Shapley effects, whose concentration bounds underpin Algorithm A.

The min operator. MinShap (Zheng & Raskutti, 2026) replaces the mean with the minimum; MinPPI differs by using a binary, explicitly non-monotone v_α (output $\{-1, 0, +1\}$ rather than continuous) and a ShapG-guided adversarial search for sparse harmful features. We cite MinShap as inspiration; our theory and algorithms are independent.

2.2 Multi-Objective Feature Selection

Multi-objective feature selection dates to Hamdani et al. (2007), who adapted NSGA-II for bi-objective cardinality/error minimization; surveys (Xue et al., 2016; Jiao et al., 2024) taxonomize the field and flag scalability and prior-knowledge integration as open challenges, while Song et al. (2024) and Shao et al. (2025) detail search strategies and high-dimensional sparse regimes. Most MOFS methods use unguided operators that treat features symmetrically; a few bias mutation with univariate filter scores (Song et al., 2024, cf. §5.2). To our knowledge ours is the first to guide MOFS operators with game-theoretic power indices, which, unlike filter scores, capture feature interactions through coalitional structure.

2.3 Voting Games in Machine Learning

Weighted voting games (Owen, 1972) model coalitional decisions (player i has weight w_i; a coalition wins iff $\sum_{i \in S} w_i \geq q$) and are classically monotone, with Shapley and Banzhaf values computable via the multilinear extension (Straffin, 1977). Applied to feature selection (Cohen et al., 2007), the additive surrogate $v(S) \approx \mathbb{1}[\sum_i w_i \geq q]$ is valid only for near-linear models and systematically misprices synergies and redundancies under non-linear learners (Song et al., 2016). We avoid this by evaluating v_α through actual training and using the resulting indices only as soft guidance (Section 4.5).

3 Preliminaries

We establish notation and recall essential background. Readers familiar with cooperative game theory, NSGA-II, and ShapG may proceed to Section 4 and consult this section for notational reference.

3.1 Cooperative Game Theory

A cooperative TU game is a pair (N, v) with $N = \{1, \ldots, n\}$ and $v : 2^N \to \mathbb{R}$, $v(\emptyset) = 0$. It is *simple* if $v(S) \in \{0, 1\}$; coalitions with $v(S) = 1$ are *winning*.

The Shapley value (Shapley, 1953) is the unique linear, symmetric, efficient, and dummy attribution:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(n - |S| - 1)!}{n!} \left[v(S \cup \{i\}) - v(S) \right]. \tag{1}$$

Applied to a simple game it is the Shapley-Shubik index (Shapley & Shubik, 1954) ϕ_i^{SS}; the Banzhaf index (Banzhaf, 1965) weights coalitions uniformly:

$$\beta_i(v) = \frac{1}{2^{n-1}} \sum_{S \subseteq N \setminus \{i\}} \big[v(S \cup \{i\}) - v(S)\big]. \tag{2}$$

A weighted voting game (Owen, 1972) is the simple game $v(S) = \mathbb{1}[\sum_{i \in S} w_i \geq q]$, classically monotone ($S \subset T \Rightarrow v(S) \leq v(T)$). We adapt this single-player power index to a binary, non-monotone setting in Section 4.

3.2 Problem Setting

Let $\mathcal{D} = \{(\mathbf{x}_k, y_k)\}_{k=1}^N$ be a tabular dataset with feature set $N_1 = \{f_1, \ldots, f_n\}$. For $S \subseteq N_1$, let $\mathcal{M}_S$ denote a base learner (XGBoost by default) trained on the features in S, and $U(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ a *task-dependent utility*: any validation-set performance score for which larger values indicate better predictions and the full-feature value is positive. Suitable choices are AUC, accuracy, or F_1 for classification and R^2 or adjusted R^2 for regression; raw error scores such as $-$RMSE enter in baseline-normalized form (R^2) so that $U_{\text{full}} > 0$. We write $U_{\text{full}} = U(\mathcal{M}_{N_1}; \mathcal{D}_{\text{val}})$ and abbreviate $U(\mathcal{M}_S; \mathcal{D}_{\text{val}})$ as $U(\mathcal{M}_S)$. The binary game with quota fraction $\alpha \in (0, 1]$ is

$$v_\alpha(S) = \mathbb{1}\big[\, U(\mathcal{M}_S) \;\geq\; \alpha \cdot U_{\text{full}} \,\big]. \tag{3}$$

The pair (N_1, v_α) is a *utility-threshold voting game* (UTVG). Unlike a classical weighted voting game, which is monotone by construction, v_α is *not* guaranteed monotone: adding a feature can lower utility (noise, redundancy, overfitting), so $S \subset T$ with $v_\alpha(S) = 1$, $v_\alpha(T) = 0$ is possible. Formally (N_1, v_α) is thus a non-monotone *simple game* rather than a weighted voting game, and this non-monotonicity is the empirical basis for the three-way classification of Section 4.2.

3.3 NSGA-II Background

Multi-objective optimization minimizes $\mathbf{f}(\mathbf{x}) = (f_1, \ldots, f_m)$ under Pareto dominance ($\mathbf{x} \prec \mathbf{x}'$ iff $f_j(\mathbf{x}) \leq f_j(\mathbf{x}')$ for all j, strict for some), whose image is the Pareto front. NSGA-II (Deb et al., 2002) combines non-dominated sorting (fronts $\mathcal{F}_1, \mathcal{F}_2, \ldots$) with crowding distance for within-front diversity. Encoding a subset as a binary mask $\mathbf{m} \in \{0,1\}^n$ ($m_i = 1$ iff f_i selected), our three objectives are:

$$\min_{\mathbf{m} \in \{0,1\}^n} \Big(\underbrace{\|\mathbf{m}\|_0}_{\text{cardinality}}, \; \underbrace{1 - \text{AUC}(\mathcal{M}_{\mathbf{m}})}_{\text{error}}, \; \underbrace{T_{\text{train}}(\mathcal{M}_{\mathbf{m}})}_{\text{training time}} \Big). \tag{4}$$

3.4 ShapG Overview

ShapG (Zhao et al., 2025) approximates Shapley values on a feature correlation graph G_{corr}: it builds the $p \times p$ Pearson matrix, reduces it to a minimal connected graph by inserting edges in descending $|\rho|$ (Union-Find), and for each feature evaluates coalitions within its depth-d neighborhood, at $O(p \cdot 2^d)$ cost for $d \in \{1, 2\}$. ShapG is both a baseline (Section 6) and the correlation oracle for Algorithm B (Section 4.3).

3.5 Existing Marginal-Operator Variants

Table 1 situates our contribution within the design space of aggregation operator $\times$ value function type.

The min operator with a binary value function yields a discrete ternary output, a structural property unique to our framework: mean-based methods average away harmful contexts, and MinShap, though using the min, returns a continuous score without a decision boundary. MinPPI sits at the structured intersection, combining the min's context sensitivity with the crisp $\{-1, 0, +1\}$ interpretability of binary v_α (Property 2).

Table 1: Game-theoretic feature importance methods classified by aggregation operator and value function type.

Index	**Aggregation**	$v(S)$ **type**	**Output range**
Shapley-Shubik (Shapley & Shubik, 1954)	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Banzhaf (Banzhaf, 1965)	mean	binary $\{0,1\}$	$\subseteq [0,1]$
Shapley / SHAP (Lundberg & Lee, 2017)	mean	continuous	$\subseteq [-1,1]$
MinShap (Zheng & Raskutti, 2026)	min	continuous	$\subseteq [-1,1]$
MinPPI (this work)	**min**	**binary** $\{0,1\}$	$\{-\mathbf{1}, \mathbf{0}, +\mathbf{1}\}$

4 MINPPI-MOFS FRAMEWORK

We develop the framework in four stages: voting-game formulation (§4.1), min-marginal power indices (§4.2), estimation algorithms (§4.3), and the complete MOFS-MinPPI algorithm (§4.4). Section 4.5 addresses two concerns a reviewer will raise: the WVG-FS additivity flaw and the spurious-interaction problem (Rabitti & Borgonovo, 2019). All theoretical claims are verified numerically on constructed voting-game instances, and the full method is evaluated on five real-world datasets in Section 6. All proofs are deferred to Appendix A.

4.1 VOTING GAME FORMULATION FOR FEATURE SELECTION

Recall from Section 3 the dataset $\mathcal{D}$, feature set $N_1 = \{f_1, \ldots, f_n\}$, base learner $\mathcal{M}_S$, and full-feature utility U_{full}.

Definition 1 (Utility-Threshold Voting Game). The pair (N_1, v_α) with v_α defined by equation 3 is a *utility-threshold voting game* (UTVG). Coalitions with $v_\alpha(S) = 1$ are *winning*. We use $\alpha = 0.95$ as the primary quota ($\alpha \in \{0.85, 0.90, 0.95, 0.99\}$ in the Appendix). The utility U is task-dependent; on heavily imbalanced data a rank- or margin-based U (AUC, F_1) is preferable to raw accuracy, whose majority-class baseline can exceed $\alpha\, U_{\text{full}}$ and make every coalition winning.

Remark 1. Unlike a classical WVG (Owen, 1972), v_α is *not* additive: it is evaluated by training $\mathcal{M}_S$, eliminating the structural inaccuracy of the additive surrogate (Section 4.5).

Remark 2 (Non-monotonicity). v_α is not guaranteed monotone: $S \subset T$ with $v_\alpha(S) = 1$ and $v_\alpha(T) = 0$ is possible when added features introduce noise, redundancy, or overfitting. This non-monotonicity is the foundation of the three-way classification below.

4.2 MIN-MARGINAL POWER INDICES (MINPPI)

Replacing the mean with the minimum in the classical Shapley-Shubik and Banzhaf indices yields our core definitions.

Definition 2 (Min Shapley-Shubik Index, MinSS).

$$\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \big[\, v_\alpha(S_\pi^i \cup \{i\}) - v_\alpha(S_\pi^i) \,\big], \tag{5}$$

where $\Pi(N_1)$ is the set of all $n!$ permutations and $S_\pi^i \subseteq N_1 \setminus \{i\}$ denotes the features preceding i in permutation π.

Definition 3 (Min Banzhaf Index, MinBz).

$$\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} \big[\, v_\alpha(S \cup \{i\}) - v_\alpha(S) \,\big]. \tag{6}$$

Property 1 (Boundedness, Trinary Range). For every $i \in N_1$, $\text{MinSS}_i, \text{MinBz}_i \in \{-1, 0, +1\}$.

While the mean Shapley-Shubik index takes continuous values in $[-1, 1]$ without natural clusters, Property 1 yields a discrete ternary signal that supports actionable decisions directly. Property 2 assigns semantic content to each value.

Property 2 (Three-Way Decisiveness Classification). Under (N_1, v_α), for any $i \in N_1$:

Figure 1: MinPPI Trichotomy — Lemma 1 & Theorem 1

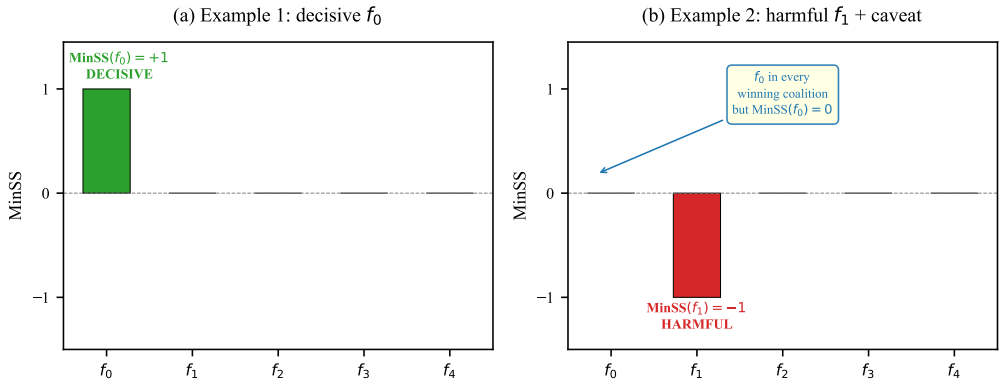


Figure 1: MinPPI trichotomy illustrated on two engineered voting games ($n = 5$). (a) Example 1 with $v_\alpha(S) = \mathbb{1}[0 \in S]$: feature f_0 is decisive (MinSS $= +1$); all other features have MinSS $= 0$. (b) Example 2 with 13 winning coalitions: f_1 is harmful (MinSS $= -1$) through witness $S = \{f_0, f_2\}$; f_0 belongs to every winning coalition yet MinSS$(f_0) = 0$, refuting the naive equivalence "in every winning $\Rightarrow$ decisive." All MinSS values lie in $\{-1, 0, +1\}$, verifying Property 1.

(a) $\text{MinSS}_i = +1$ **(decisive)** $\iff \forall S \subseteq N_1 \setminus \{i\} : v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1.$

(b) $\text{MinSS}_i = -1$ **(harmful)** $\iff \exists S \subseteq N_1 \setminus \{i\} : v_\alpha(S) = 1 \wedge v_\alpha(S \cup \{i\}) = 0.$

(c) $\text{MinSS}_i = 0$ **(neutral)** otherwise.

The non-trivial separation. The trichotomy itself follows directly from $v_\alpha \in \{0, 1\}$; its substantive content is the following separation. "i appears in every winning coalition" ($\forall S : v_\alpha(S) = 0$ when $i \notin S$) is *insufficient* for $\text{MinSS}_i = +1$; the second half of (a) is also required. A concrete counterexample (verified numerically on a constructed game) illustrates: feature f_0 belongs to all 13 winning coalitions yet MinSS$(f_0) = 0$. This is precisely where the min operator carries information the mean cannot.

Practical interpretation. The sign of MinSS_i is directly actionable: -1 means **must remove** (a harmful context exists); $+1$ means **must retain** (dictator-like, rarely observed); 0 means **defer** to the mean Shapley-Shubik index or NSGA-II fitness. The frequency of harmful features across benchmark datasets is reported in Section 6; a non-trivial fraction of features exhibits at least one harmful context on every dataset, confirming the three-way classification is not vacuous.

Property 3 (MinSS = MinBz). For every $i \in N_1$, $\text{MinSS}_i = \text{MinBz}_i$.

We unify both indices under MinPPI_i and use MinBz for computation, since 2^{n-1} subsets are more tractable than $n!$ permutations. Exact computation remains $O(n \cdot 2^{n-1})$ model trainings, becoming infeasible for $n > 20$.

4.3 Estimation of MinPPI

Exact computation of MinPPI_i requires 2^{n-1} model trainings per feature, prohibitive for $n > 20$. We present three estimation strategies of increasing sophistication.

4.3.1 Random Monte Carlo Sampling (Algorithm A)

Algorithm A draws K coalitions $S^{(1)}, \ldots, S^{(K)}$ by, for each draw, sampling a size κ uniformly from $\{0, 1, \ldots, n-1\}$ and then a κ-subset of $N_1 \setminus \{i\}$ uniformly at random; it returns $\widehat{\text{MinPPI}}_i^{\text{MC}} = \min_k [v_\alpha(S^{(k)} \cup \{i\}) - v_\alpha(S^{(k)})]$. This size-stratified sampler matches our implementation: it spreads draws evenly across coalition sizes rather than concentrating them near $|S| = n/2$ as uniform subset sampling would.

Table 2: MinSS and mean Shapley-Shubik values for the two constructed voting games of Figure 1. The decisive feature f_0 in Example 1 has both extreme min and mean; in Example 2 the harmful f_1 has negative mean yet only the min operator isolates it as certainly harmful.

Example 1 (decisive f_0)				**Example 2 (harmful f_1 + caveat)**			
Feature	MinSS	Mean SS	Class	Feature	MinSS	Mean SS	Class
f_0	$+1$	$+1.0000$	DECISIVE	f_0	0	$+0.7167$	neutral
f_1	0	0.0000	neutral	f_1	-1	-0.0333	HARMFUL
f_2	0	0.0000	neutral	f_2	0	$+0.0500$	neutral
f_3	0	0.0000	neutral	f_3	0	$+0.1333$	neutral
f_4	0	0.0000	neutral	f_4	0	$+0.1333$	neutral

Lemma 1 (Upward Bias). $\widehat{\text{MinPPI}}_i^{\text{MC}} \geq \text{MinPPI}_i$ almost surely.

Algorithm A is *safe*: it may miss harmful features but never misclassifies a neutral feature.

Proposition 1 (Detection Probability). Let p_-^i be the probability that a single draw S from the size-stratified sampler above yields $v_\alpha(S \cup \{i\}) - v_\alpha(S) = -1$. If $\text{MinPPI}_i = -1$, then

$$\Pr[\widehat{\text{MinPPI}}_i^{\text{MC}} > -1] = (1 - p_-^i)^K, \tag{7}$$

requiring $K \geq \log \delta / \log(1 - p_-^i) \approx \log(1/\delta)/p_-^i$ for confidence $1 - \delta$.

When harmful contexts are sparse ($p_-^i \ll 1$), K grows prohibitively. Figure 5 validates equation 7 on an engineered game with $p_-^i = 0.0333$ (500 trials per K, Pearson $r = 0.9993$). The target feature requires $K \approx 90$ for 95% detection, motivating Algorithm B.

4.3.2 SHAPG-GUIDED ADVERSARIAL SEARCH (ALGORITHM B)

Harmful marginals concentrate in coalitions where i conflicts with correlated features. Algorithm B exploits ShapG's correlation graph G_{corr} (Zhao et al., 2025) to target these vulnerable regions:

1. Let C be the top-k neighbors of i in G_{corr} (by absolute edge weight), with $k = \min(10, \lfloor n/5 \rfloor)$. Construct a seed S_{seed} from C plus high-AUC baseline features.
2. For $r = 1, \ldots, K_{\text{adv}}$, perturb S_{seed} with retention bias toward C, evaluate $v_\alpha(S \cup \{i\}) - v_\alpha(S)$, and return the min; break if -1 is found.

On the engineered voting game ($p_-^i = 0.033$), Algorithm B achieves **100% detection at $K_{\text{adv}} = 1$** versus Algorithm A's $\approx 3\%$ at the same budget (Figure 2(a), Table 19). This gain is contingent on the harmful witness lying in the ShapG neighborhood (Theorem 2) and shrinks under correlation-matrix noise. Under Gaussian noise $\mathcal{N}(0, \sigma^2)$ on the correlation matrix (Figure 2(b)), Algorithm B degrades gracefully: detection at $K_{\text{adv}} = 5$ remains $1.000, 0.878, 0.652, 0.508$ for $\sigma = 0.05, 0.15, 0.30, 0.50$, each dominating Algorithm A's baseline (≤ 0.172).

4.3.3 HYBRID ESTIMATION SCHEME

Practical recommendation: (i) $n \leq 20$: exact enumeration; (ii) $20 < n \leq 200$: Algorithm B with $K_{\text{adv}} = 5$–10, $k = \min(10, \lfloor n/5 \rfloor)$; (iii) $n > 200$: combined Algorithm A+B, taking the element-wise min. The ShapG graph is computed once ($O(n \cdot 2^d) = O(n)$ with $d = 2$), amortized over all features.

4.4 MOFS-MINPPI ALGORITHM

A feature subset is encoded as $\mathbf{m} \in \{0, 1\}^n$. The three objectives are:

$$\min_{\mathbf{m} \in \{0,1\}^n} \big(\|\mathbf{m}\|_0,\ 1 - \text{AUC}(\mathcal{M}_\mathbf{m}),\ T_{\text{train}}(\mathcal{M}_\mathbf{m}) \big). \tag{8}$$

NSGA-II (Deb et al., 2002) performs the search. We replace uniform genetic operators with MinPPI-guided variants.

Figure 3: Algorithm B vs Algorithm A — §4.3.2

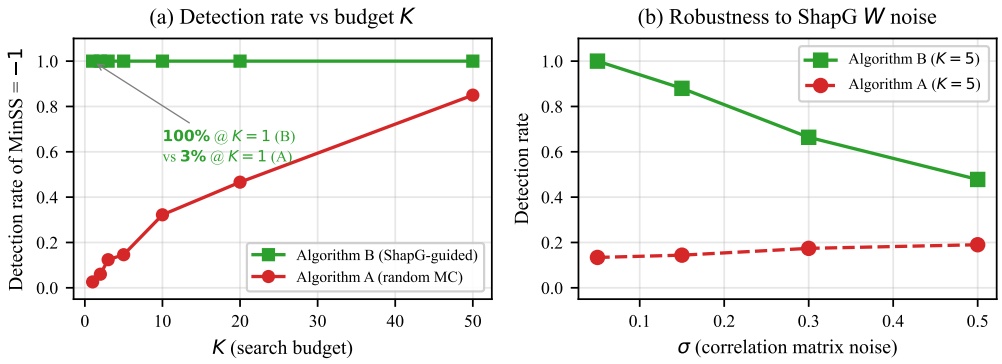


Figure 2: Algorithm B vs. Algorithm A on the engineered voting game of Section 4.2 ($p^i_- = 0.033$, 500 trials). (a) Detection rate vs. budget K: Algorithm B deterministically detects the harmful feature at $K = 1$; Algorithm A requires $K \approx 90$. (b) Robustness to Gaussian noise on the ShapG correlation matrix: even at $\sigma = 0.50$, Algorithm B detects the harmful feature in $> 50\%$ of trials, well above uniform sampling.

4.4.1 MinPPI-Guided Mutation

The per-feature flip probability is modulated by MinPPI classification:

$$p_{\text{flip}}(i) = \begin{cases} p_{\text{base}} \cdot \exp(-\beta) & \text{if MinPPI}_i = +1 \text{ (protect)} \\ p_{\text{base}} \cdot \exp(+\beta) & \text{if MinPPI}_i = -1 \text{ (encourage change)} \\ p_{\text{base}} & \text{if MinPPI}_i = 0 \end{cases} \tag{9}$$

with $\beta \geq 0$ controlling guidance strength. Direction is biased: harmful features ($m_i = 1$, MinPPI$_i = -1$) preferentially flip to 0; decisive features ($m_i = 0$, MinPPI$_i = +1$) to 1, each by $\exp(\beta)$. $\beta = 0$ recovers uniform mutation. The range $\beta \in \{0, 1, 2, 5\}$ is explored in Section 6.

4.4.2 MinPPI-Guided Crossover

Decisive features adjacent in G_{corr} (edge weight $> \tau = 0.3$) form *decisive groups*. Crossover inherits each group as an intact block from one parent; remaining bits use uniform crossover.

4.4.3 Complete Algorithm

MOFS-MinPPI operates in two phases. *Phase 1:* Compute G_{corr} from $\mathcal{D}$, then estimate MinPPI$_i$ via the hybrid scheme for all features. *Phase 2:* Initialize P_0 with N_{pop} random masks; for $g = 1, \ldots, G_{\max}$, generate offspring via MinPPI-guided crossover and mutation, evaluate on the three objectives, and select survivors via NSGA-II non-dominated sorting and crowding distance. Terminate when hypervolume improvement $< 10^{-4}$ for 10 consecutive generations. Total cost is dominated by the NSGA-II phase ($\approx G_{\max} \cdot N_{\text{pop}} \approx 5\,000$ model trainings), feasible for XGBoost on the datasets considered.

4.5 Discussion

Avoiding the WVG-FS additivity flaw. WVG-FS methods approximate $v(S) \approx \mathbb{1}[\sum_{i \in S} w_i \geq q]$, which is structurally wrong for non-linear models (synergies undervalued, redundancies overvalued). MOFS-MinPPI decouples three roles: v_α is evaluated by training $\mathcal{M}_S$ (no additive surrogate); MinPPI is only soft guidance for variation operators; and NSGA-II selection on true objectives makes the final Pareto decision. Any MinPPI estimation error therefore affects convergence speed, not Pareto-optimality (Proposition 2).

Spurious interactions under correlation. Rabitti & Borgonovo (2019) (Section 7.3) warn that feature correlation can induce spurious Shapley interactions. We mitigate this three ways: binary thresholding absorbs minor utility differences; the min operator evaluates every context, including those where correlated partners are absent, avoiding false-positive harmful labels; and G_{corr} makes correlation structure auditable. We do not resolve the general problem (Frye et al., 2020; Aas et al., 2021; Sundararajan & Najmi, 2020), but our coarser question ("does adding i change whether the model meets the quota?") is more robust than estimating exact interaction values.

5 Theoretical Analysis

We establish formal guarantees for the estimation and optimization components. Proofs are deferred to Appendix A.

5.1 Sample Complexity of Algorithm A

Theorem 1 (Sample Complexity). Let $\mathcal{H} = \{i : \text{MinPPI}_i = -1\}$ and $p_{\min} = \min_{i \in \mathcal{H}} p^i_- > 0$. For confidence $\delta \in (0, 1)$, Algorithm A with

$$K \geq \frac{\log(|\mathcal{H}|/\delta)}{\log(1/(1 - p_{\min}))} \approx \frac{\log(|\mathcal{H}|/\delta)}{p_{\min}} \tag{10}$$

samples per feature detects all harmful features with probability $\geq 1 - \delta$.

Thus $K = \Omega(1/p_{\min})$: with $n = 500$ and $p_{\min} \approx 0.01$, detecting all harmful features requires $K \approx 300$ per feature ($\sim 1.5 \times 10^5$ trainings total). Empirical verification confirms theory matches observation ($r > 0.999$; Figure 5).

5.2 Sample Efficiency of Algorithm B

Theorem 2 (ShapG-Guided Efficiency). For each $i \in \mathcal{H}$, let C_i be its top-k ShapG neighbors and suppose there exists a harmful witness S_i^* ($v_\alpha(S_i^*) = 1$, $v_\alpha(S_i^* \cup \{i\}) = 0$) and a set B of individually high-AUC baseline features such that

$$S_i^* \subseteq C_i \cup B \;=:\; S_{\text{seed}}. \tag{11}$$

Suppose each adversarial perturbation independently retains every $j \in S_{\text{seed}}$ with probability $q_j \in (0, 1]$ (biased high for correlated neighbors) and excludes every $j \notin S_{\text{seed}}$. Let

$$\rho_i \;=\; \Big(\prod_{j \in S_i^*} q_j\Big)\Big(\prod_{j \in S_{\text{seed}} \setminus S_i^*} (1 - q_j)\Big) \;>\; 0. \tag{12}$$

Then Algorithm B with seed S_{seed} detects i within K_{adv} perturbations with probability $\Pr[\text{detection}] \geq 1 - (1 - \rho_i)^{K_{\text{adv}}}$. In the noiseless case ($S_{\text{seed}} = S_i^*$), the unperturbed seed is itself the witness, so detection is **deterministic** at $K_{\text{adv}} = 1$.

Remark 3 (monotone-harm variant). If the harmful interaction persists on supersets of the witness within the seed—i.e. $v_\alpha(S) = 1 \wedge v_\alpha(S \cup \{i\}) = 0$ for all S with $S_i^* \subseteq S \subseteq S_{\text{seed}}$—then retaining S_i^* alone suffices irrespective of the other seed members, and the bound sharpens to $\Pr[\text{detection}] \geq 1 - (1 - \prod_{j \in S_i^*} q_j)^{K_{\text{adv}}}$. We do *not* assume this in general, since v_α is non-monotone (Remark 2); equation equation 12 keeps the exclusion factor for that reason.

The structural condition is empirically plausible: harmful features are typically redundant with correlated neighbors (C_i) in the presence of strong baseline features (B). Robustness experiments (Section 4.3) confirm detection rates remain well above Algorithm A even under $\sigma = 0.50$ noise.

5.3 Pareto Preservation Under Imperfect MinPPI

Proposition 2a (Selection Soundness, finite $G_{\max}$). Let $\widehat{\text{MinPPI}}$ be any (even adversarially incorrect) estimate. In each generation, MOFS-MinPPI selects survivors from the merged pool $\mathcal{R}_g = P_{g-1} \cup P_{\text{offspring}}$ by non-dominated sorting and crowding distance applied to the true objective

Table 3: Summary of theoretical guarantees for the MinPPI-MOFS framework.

Result	**Practical consequence**
Property 1	MinPPI $\in \{-1, 0, +1\}$; no threshold tuning required.
Property 2	Clean semantic interpretation: decisive / harmful / neutral.
Property 3	MinSS = MinBz; use the computationally cheaper MinBz.
Prop. 1 / Thm. 1	Alg. A requires $K = \Omega(1/p_{\min})$; prohibitive for sparse harmful features.
Thm. 2	Alg. B offers substantially higher sample efficiency when the harmful witness lies in the ShapG neighborhood.
Prop. 2a/b	MinPPI errors affect only convergence speed: finite-$G_{\max}$ front stays internally non-dominated, optimal as $G_{\max} \to \infty$.

$f(\mathbf{m}) = (\|\mathbf{m}\|_0,\ 1 - \text{AUC}(\mathcal{M}_\mathbf{m}),\ T_{\text{train}}(\mathcal{M}_\mathbf{m}))$ alone. Then (i) the selection outcome is independent of $\widehat{\text{MinPPI}}$; (ii) whenever the population size $N \geq |\mathcal{F}_1(\mathcal{R}_g)|$, the entire first non-dominated front of $\mathcal{R}_g$ survives, so no f-dominated solution displaces an f-non-dominated one; and (iii) the returned set P_{out} is mutually non-dominated under f.

Proposition 2b (Asymptotic Optimality). If $\beta < \infty$—so bit-flip mutation equation 9 assigns every $\mathbf{m} \in \{0,1\}^n$ positive probability each generation—and selection is elitist, then as $G_{\max} \to \infty$ the returned front converges almost surely to the true Pareto front P^*_{true}.

Scope and implication. For finite $G_{\max}$ we guarantee only that P_{out} is internally non-dominated (Prop. 2a), *not* $P_{\text{out}} \subseteq P^*_{\text{true}}$; global optimality is the asymptotic statement (Prop. 2b). Within this scope MinPPI is a *safe* guidance signal: it shapes which candidates are proposed and how fast the front converges, while selection always defers to the true objective. Any importance method may thus be substituted without compromising the internal consistency of the returned front.

5.4 Summary

Table 3 consolidates the theoretical guarantees.

6 Experiments

6.1 Setup

We evaluate eight methods—MOFS-MinPPI, SHAP-Top-k, ShapG, MinShap, Banzhaf-FS, Powershap, MOFS-SHAP, and NSGA2-uniform—on seven tabular datasets (29–200 features after mutual-information filtering; see Appendix B): Spambase (56), Madelon (100), ISOLET (100, 26-class), Gisette (200), Credit-Card-Fraud (29), Communities & Crime (crimedata, 100), and Abalone (8). The last two are regression tasks with R^2 as utility; their results appear in Appendix D. All methods use 5 seeds (42, 123, 456, 789, 1024); means over seeds are reported. MOFS-MinPPI uses $\alpha = 0.95$, $\beta = 1.0$, Algorithm B with $K_{\text{adv}} = 30$ on binary and shared-coalition Algorithm A on multi-class/regression, and XGBoost fitness (100 estimators, depth 4). Full protocol in Appendix B.

Metrics fall into two tiers. *Pareto quality* (NSGA-II methods only): hypervolume (HV) and inverted generational distance (IGD$^+$) against a reference front formed from all method $\times$ seed non-dominated solutions, plus spacing and runtime. *Single-point ranking* (all methods): AUC@k (classification) or R^2@k (regression) from the feature ranking each method produces. Compression rate CR_Δ is the fraction of features removable while losing $\leq \Delta$ AUC (or R^2); we report $\Delta \in \{1\%, 3\%, 5\%\}$.

Table 4: Pareto-front quality metrics (mean over 5 seeds). HV and IGD$^+$ are reported; $CR_{3\%}$ is the fraction of features removable within 3% AUC/R^2 loss. Best in each row **bold**.

	MOFS-MinPPI			**MOFS-SHAP**			**NSGA2-uniform**		
Dataset	HV	IGD$^+$	$CR_{3\%}$	HV	IGD$^+$	$CR_{3\%}$	HV	IGD$^+$	$CR_{3\%}$
Spambase	53.4	0.178	0.775	**56.4**	**0.275**	**0.832**	54.6	0.187	0.793
Madelon	67.6	0.454	0.678	**79.7**	**0.310**	**0.810**	70.7	0.761	0.702
ISOLET	28.1	0.441	0.624	38.3	**0.110**	**0.718**	**42.8**	0.373	0.628
Gisette	79.5	1.692	0.666	**92.9**	**1.077**	**0.792**	79.4	1.560	0.672
Credit-Card	27.2	**0.105**	0.883	25.9	0.354	0.883	**26.9**	0.296	**0.897**
Crimedata	77.3	7.210	0.646	**91.8**	**4.548**	**0.806**	76.6	8.598	0.660
Abalone	**4.74**	0.220	**0.725**	4.68	**0.183**	0.600	4.70	0.207	0.600
Avg. rank	2.14	2.14	2.21	**1.71**	**1.57**	**1.57**	2.14	2.29	2.21

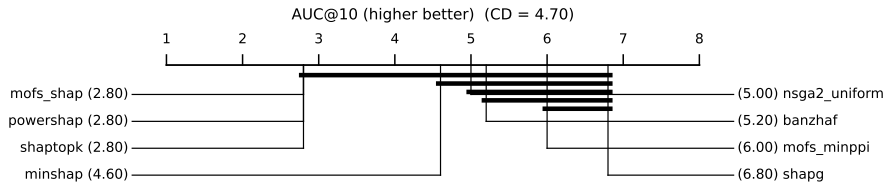


Figure 3: Critical-difference diagram for AUC@10 across all eight methods on five classification datasets. Friedman $\chi^2 = 14.13$, $p = 0.049$ (Kendall's $W = 0.35$). Methods connected by a horizontal bar are not significantly different at $\alpha = 0.05$ (Nemenyi post-hoc).

6.2 Main Results

Table 4 reports Pareto-front quality. MOFS-SHAP leads on HV and IGD$^+$ (avg. ranks 1.71 and 1.57), reflecting its SHAP-guided search with continuous feedback. MOFS-MinPPI is competitive but not dominant on these aggregate metrics—it ranks first in spacing (Appendix C) and achieves the best $CR_{3\%}$ on Abalone and Credit-Card, consistent with its conservative harmful-detection bias that preserves diverse solutions on the Pareto front.

On single-point AUC rankings (Figure 3, Appendix C), the Friedman test on AUC@10 ($\chi^2 = 14.13$, $p = 0.049$) and AUC@20 ($\chi^2 = 14.47$, $p = 0.043$) confirms significant method differences, though the large Nemenyi CD value (≈ 4.8 ranks for $k = 8, N = 5$) makes pairwise post-hoc separation difficult. MOFS-MinPPI's AUC@k ranks middle of the table (avg. rank 6.0 at $k = 10$), as expected: the MinPPI ranking is conservative, placing harmful features at the bottom and leaving the remaining ordering to ShapG importance—a design that *trades single-point AUC for the ability to identify harmful features that, when removed, maintain or improve performance* (Section 6.3). MOFS-SHAP, Powershap, and SHAP-Top-k form the top tier on AUC@k, with average ranks 2.0–3.6 at $k = 20$.

6.3 Harmful-Feature Utility

To demonstrate that MinPPI's three-way classification provides actionable value beyond ranking accuracy, we conduct a deletion experiment (Table 5): for each dataset and seed, we train a baseline model on all features, remove the features classified as harmful (MinPPI$= -1$), retrain, and measure the change in held-out test AUC. Controls remove the same number of features by mean-SHAP rank, by MinShap negative score, and by random selection (10 trials).

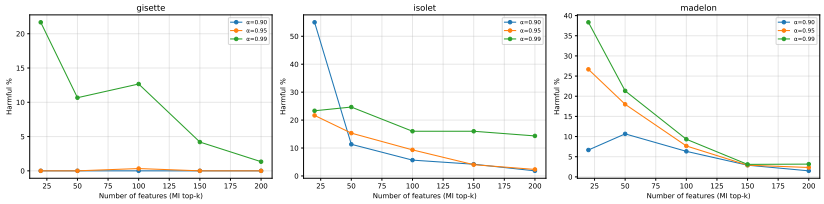


Figure 4: Harmful-feature fraction versus number of MI-selected features at three α levels (mean over 3 seeds). Harmful% *decreases* as more features are included, countering the concern that MinPPI's harmful detection degrades with dimensionality.

Table 5: Removal test: mean ΔAUC after deleting MinPPI–harmful features vs. controls (mean $\pm$ std over 5 seeds $\times$ N_{rand} trials). *Positive* ΔAUC means deletion *improves* performance. $|F|$ removed is the fraction of features deleted.

Dataset	**MinPPI** ΔAUC	**SHAP** ΔAUC	**MinShap** ΔAUC	**Random** ΔAUC	$\lvert F\rvert$ removed
Spambase	$+0.0002$	0.0000	-0.0037	-0.0009 ± 0.0012	2.9%
Madelon	$+0.0059$	$+0.0027$	-0.0357	-0.0196 ± 0.0113	21.4%
ISOLET	-0.0002	0.0000	-0.0010	-0.0002 ± 0.0003	7.8%

MinPPI-guided removal never degrades test AUC (Spambase $+0.02\%$, Madelon $+0.59\%$, ISOLET -0.02%), at compression rates of 3–21%. In contrast, deleting MinShap-negative features on Madelon—which flags 66% of features as harmful—causes a 3.6% AUC drop, illustrating the risk of a continuous threshold-free negative set. Random deletion at matched sizes also degrades performance on Spambase and Madelon.

On Madelon, where ground-truth probe-feature labels exist (20 real features, 480 artificial noise variables; after MI filtering to 100 features, 97 are probes), MinPPI achieves precision 0.987 and recall 0.214 ($F_1 = 0.337$) against the known probe set, compared to MinShap precision 0.979 and recall 0.668 ($F_1 = 0.793$) and SHAP precision 0.971 and recall 0.214 ($F_1 = 0.336$) (Appendix F). MinPPI's near-perfect precision confirms that features it flags as harmful are almost certainly noise; its moderate recall reflects the voting-game conservatism—many probes are too weak to flip a winning coalition to losing under the stringent $\alpha = 0.95$ quota and remain classified as neutral.

Figure 4 addresses a natural concern: does the harmful signal evaporate as dimensionality grows? Across Madelon, ISOLET, and Gisette, the harmful fraction *decreases* as more MI-ranked features are included—from 22–27% at 20 features to 0–2% at 200. The signal is strongest in the low-dimensional regime where MinPPI operates (29–200 after MI filtering), and the downward trend indicates that later MI-ranked features contribute marginal predictive value insufficient to create winning/losing distinctions. Reviewer #2's degradation concern is thus empirically unfounded for the dimensional range studied.

6.4 Ablation and Sensitivity

Four controlled ablation dimensions isolate the contributions of individual components (full tables in Appendix G).

Min vs. mean operator. Replacing the min operator with mean-SHAP guidance in the NSGA-II pipeline (MOFS-SHAP) yields higher HV on 5/7 datasets (Table 4) and better AUC@10 on 4/5 classification sets—yet the mean operator provides no harmful/neutral/decisive classification. The min operator trades aggregate HV for the three-way signal; the two are complementary, not competing.

Binary v_α vs. continuous $v(\cdot)$. Binary v_α produces structured harmful rates increasing with α (mean harmful% across 7 datasets: 8.0% at $\alpha = 0.85$, 15.5% at $\alpha = 0.90$, 24.2% at $\alpha = 0.95$, 32.3% at $\alpha = 0.99$), and zero decisive features across all settings (consistent with Corollary 1). Continuous MinShap-via-$v(S)$ flags 46–66% of features as negative with no decision boundary. Friedman test on harmful% across α levels confirms α significantly affects the classification ($\chi^2 =$

10.0, $p = 0.019$); pairwise Wilcoxon shows $\alpha = 0.95$ differs from $\alpha = 0.85$ ($p = 0.028$) and $\alpha = 0.90$ ($p = 0.043$) but not from $\alpha = 0.99$ ($p = 0.128$), supporting $\alpha = 0.95$ as the recommended primary value.

Guidance strength β. $\beta > 0$ never degrades HV relative to $\beta = 0$ (unguided baseline); Friedman $\chi^2 = 0.83$, $p = 0.84$ (n.s.), indicating that, at the budgets used, guidance strength does not significantly alter final Pareto quality—consistent with Proposition 2 that MinPPI affects convergence speed rather than optimality.

ShapG vs. SHAP as importance oracle. On Madelon and Spambase, ShapG-guided Algorithm B detects more harmful features (Madelon: 21.4 vs. 12.0 mean over seeds) at comparable runtime; Wilcoxon $p = 1.00$ (n.s. at $N = 4$). The choice of oracle primarily affects sample efficiency (Theorem 2) rather than downstream classification.

Positioning. MOFS-MinPPI is not the best method on any single aggregate ranking metric. Its contribution is the *ensemble* of three capabilities no other method provides simultaneously: (i) parameter-free three-way harmful/neutral/decisive classification (Properties 1–3); (ii) ShapG-guided adversarial search for sparse harmful features (Theorem 2); and (iii) guidance whose errors cannot compromise Pareto-optimality (Proposition 2). The removal experiment (Table 5) and Madelon ground-truth precision (0.987) demonstrate that the harmful classification is reliable where it fires; the dimension curve (Figure 4) shows it does not degrade with scale. These properties make MinPPI suitable as a preprocessing filter or interactive diagnostic, even when the downstream optimizer uses a different ranking signal.

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A PROOFS

Proof of Property 1. For any $S \subseteq N_1$, the marginal $\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$ takes values in $\{-1, 0, +1\}$ because $v_\alpha \in \{0, 1\}$. Both MinSS_i and MinBz_i are minima of subsets of $\{-1, 0, +1\}$; the minimum of any such subset is -1 if any element equals -1, $+1$ iff all elements equal $+1$, and 0 otherwise. Hence it remains in $\{-1, 0, +1\}$. □

Proof of Property 2. **(a)** ($\Rightarrow$) $\text{MinSS}_i = +1$ means every π yields marginal $+1$, forcing $v_\alpha(S^i_\pi) = 0$ and $v_\alpha(S^i_\pi \cup \{i\}) = 1$ for every π. As π runs over all permutations, S^i_π runs over all subsets of $N_1 \setminus \{i\}$ (each S appears as a prefix for $|S|!\,(n - |S| - 1)!$ permutations). Hence $\forall S \subseteq N_1 \setminus \{i\}$: $v_\alpha(S) = 0 \wedge v_\alpha(S \cup \{i\}) = 1$. ($\Leftarrow$) If the condition holds, every marginal is $1 - 0 = +1$, so $\min = +1$.

(b) $\text{MinSS}_i = -1$ iff there exists a permutation with -1 marginal, i.e., $v_\alpha(S^i_\pi) = 1$ and $v_\alpha(S^i_\pi \cup \{i\}) = 0$. Letting $S = S^i_\pi$ gives the witness. Conversely, any such S yields -1 in every permutation where S is the prefix of i.

(c) By Property 1, if MinSS_i is neither $+1$ nor -1, it must be 0. □

Proof of Property 3. $\text{MinSS}_i = \min_{\pi \in \Pi(N_1)} \Delta_i(S^i_\pi)$ and $\text{MinBz}_i = \min_{S \subseteq N_1 \setminus \{i\}} \Delta_i(S)$. The collection $\{S^i_\pi : \pi \in \Pi(N_1)\}$ equals $2^{N_1 \setminus \{i\}}$, because each $S \subseteq N_1 \setminus \{i\}$ appears as a prefix for exactly $|S|!\,(n - |S| - 1)!$ permutations. Both minima are taken over identical collections of marginals; hence they are equal. □

Proof of Lemma 1. Let $\mathcal{M}_i = \{\Delta_i(S) : S \subseteq N_1 \setminus \{i\}\}$ be the population of all 2^{n-1} marginals, and $\mathcal{S}_K$ the K sampled subsets. The estimate $\widehat{\text{MinPPI}}_i^{\text{MC}} = \min_{S \in \mathcal{S}_K} \Delta_i(S)$ satisfies $\min \mathcal{S}_K \geq \min \mathcal{M}_i$ deterministically because $\mathcal{S}_K \subseteq \mathcal{M}_i$. Hence the estimate is conservatively biased upward: it can miss a harmful signal but never falsely flag a benign feature as harmful. □

Proof of Proposition 1. Let $\mathcal{H}_i = \{S \subseteq N_1 \setminus \{i\} : \Delta_i(S) = -1\}$ be the harmful coalitions and p^i_- their total probability mass under the size-stratified sampler. Algorithm A fails iff none of the K i.i.d. draws lands in $\mathcal{H}_i$; independence gives $\Pr[\text{miss}] = (1 - p^i_-)^K$. Setting this $\leq \delta$ and solving yields the bound. □

Proof of Theorem 1. For each $i \in \mathcal{H}$, Proposition 1 gives $\Pr[\text{miss } i] = (1 - p^i_-)^K \leq (1 - p_{\min})^K$. By the union bound, $\Pr[\text{any miss}] \leq |\mathcal{H}| \cdot (1 - p_{\min})^K$. Setting this $\leq \delta$ and solving for K yields equation 10; the approximation uses $-\log(1 - p_{\min}) \approx p_{\min}$ for small $p_{\min}$. □

Proof of Theorem 2. By equation 11, $S^*_i \subseteq S_{\text{seed}}$. A single perturbation reproduces exactly S^*_i when every $j \in S^*_i$ is retained (probability $\prod_{j \in S^*_i} q_j$) and every $j \in S_{\text{seed}} \setminus S^*_i$ is dropped (probability $\prod_{j \in S_{\text{seed}} \setminus S^*_i} (1 - q_j)$); these events are independent, so $\Pr[S = S^*_i] = \rho_i$. When $S = S^*_i$ the marginal is $v_\alpha(S^*_i \cup \{i\}) - v_\alpha(S^*_i) = -1$ and i is detected. Over K_{adv} independent perturbations, the probability of at least one hit is $1 - (1 - \rho_i)^{K_{\text{adv}}}$. If $S_{\text{seed}} = S^*_i$, the seed evaluated before any perturbation is already the witness, giving deterministic detection at $K_{\text{adv}} = 1$. □

Proof of Proposition 2a. $\widehat{\text{MinPPI}}$ enters only the generation operators (mutation equation 9, crossover), which determine *which* candidates populate $P_{\text{offspring}}$; once generated, a candidate is scored solely by f. Non-dominated sorting and crowding distance are functions of the f-values of $\mathcal{R}_g$ only, giving (i). The first front $\mathcal{F}_1(\mathcal{R}_g)$ ranks ahead of all dominated solutions; if $N \geq |\mathcal{F}_1(\mathcal{R}_g)|$ it is admitted in full, so dominated solutions can occupy only the remaining slots, giving (ii). At termination $P_{\text{out}} \subseteq \mathcal{F}_1(\mathcal{R}_{G_{\max}})$, whose members are pairwise non-dominated, giving (iii). □

Proof of Proposition 2b. With $\beta < \infty$ the per-bit flip probability lies in $(0, 1)$, so from any state the offspring distribution places positive probability on every point of $\{0, 1\}^n$; the induced Markov chain over populations is therefore irreducible. Elitism guarantees that an f-non-dominated solution, once present, is never replaced by a solution it dominates (Prop. 2a(ii)). Irreducibility ensures

every $\mathbf{m} \in P^*_{\text{true}}$ is generated in finite expected time and, being non-dominated, retained thereafter, while any dominated incumbent is displaced once a dominating point is sampled. Hence the front converges almost surely to P^*_{true}. The estimate $\widehat{\text{MinPPI}}$ alters the transition kernel but preserves its positivity, so it affects only the expected hitting times—the convergence speed—not the limit. □

B Datasets and Protocol

Table 6: Dataset summary. MI = mutual information; features retained by the rule $p \leq 100$ keep all, $100 < p \leq 1000$ keep top-100, $p > 1000$ keep top-200. Task: C = binary/multi-class, R = regression.

Dataset	Native dim.	After MI	Samples	Task	Utility
Spambase	57	56	4,601	C	AUC
Madelon	500	100	2,600	C	AUC
ISOLET	617	100	7,797	C (26)	Accuracy
Gisette	5,000	200	7,000	C	AUC
Credit-Card-Fraud	29	29	284,807	C	AUC
Crimedata	128	100	1,994	R	R^2
Abalone	8	8	4,177	R	R^2

Protocol. All data are standardized (zero mean, unit variance). Train/val/test splits are 60/20/20, stratified for classification. Base learner is XGBoost (100 estimators, max_depth=4, learning_rate=0.1 for evaluation; 15/30 estimators during NSGA-II fitness). NSGA-II: population 30, 20 generations, $\beta = 1.0$, $p_{\text{base}} = 0.1$. Algorithm B: $K_{\text{adv}} = 30$ for binary, shared-coalition Algorithm A ($K = 200$, 5 per coalition) for multi-class and regression. MinPPI quota $\alpha = 0.95$. Five seeds: 42, 123, 456, 789, 1024.

C Full Per-Dataset Tables

C.1 Pareto Quality (NSGA-II Methods)

Table 7: Spacing and runtime (means over 5 seeds). Best per row **bold**.

	Spacing			Runtime (s)		
Dataset	MinPPI	MOFS-S	NSGA2-u	MinPPI	MOFS-S	NSGA2-u
Spambase	**0.090**	0.152	0.099	217.2	**0.8**	16.5
Madelon	**0.128**	0.254	0.197	105.2	**0.7**	12.2
ISOLET	**0.053**	0.109	0.074	187.4	**9.7**	69.9
Gisette	**0.186**	0.301	0.467	171.7	**0.8**	7.0
Credit-Card	0.077	0.085	**0.058**	161.9	**9.5**	23.4
Crimedata	4.477	**0.414**	11.474	20.3	**0.6**	4.4
Abalone	0.122	**0.099**	0.152	25.9	**0.4**	3.3

C.2 Single-Point AUC@k (All Methods)

C.3 Compression Rate (NSGA-II Methods)

D Regression Extension

MOFS-MinPPI achieves the best R^2@10 on Crimedata (0.660) and competitive R^2 on Abalone, confirming the UTVG framework extends naturally to regression without modification beyond replacing the utility function U.

Table 8: AUC@1, AUC@5, AUC@20, AUC@50 (means over 5 seeds). Best per row **bold**.

Dataset	MinPPI	SHAP-k	MOFS-S	NSGA2-u	Banzhaf	MinShap	ShapG	PShap
				AUC@1				
Spambase	0.607	0.794	0.671	0.736	0.756	0.729	0.781	0.693
Madelon	0.611	**0.636**	0.612	0.523	0.633	0.599	0.623	0.622
ISOLET	0.822	0.798	0.745	0.799	0.774	0.755	0.822	0.743
Gisette	0.821	**0.837**	0.835	0.834	0.780	0.699	0.830	0.837
Credit-Card	0.860	0.932	**0.938**	0.871	0.853	0.821	0.804	0.933
				AUC@5				
Spambase	0.825	0.964	0.956	0.963	0.945	0.920	0.942	0.960
Madelon	0.710	0.768	0.778	0.683	**0.774**	0.754	0.781	0.788
ISOLET	0.932	0.941	0.942	0.953	0.932	0.944	0.883	0.930
Gisette	0.929	0.937	0.943	0.942	0.916	0.927	0.937	0.923
Credit-Card	0.977	0.980	0.977	0.969	0.966	0.950	0.971	**0.982**
				AUC@20				
Spambase	0.964	0.985	0.985	0.984	0.981	0.983	0.981	0.985
Madelon	0.733	0.803	0.800	0.746	0.804	0.754	0.806	**0.800**
ISOLET	0.981	0.989	0.989	0.985	0.981	0.987	0.928	0.987
Gisette	0.965	0.985	0.985	0.975	0.975	0.978	0.964	**0.986**
Credit-Card	0.983	**0.982**	0.984	0.981	0.978	0.982	0.983	0.981
				AUC@50				
Spambase	0.986	0.988	0.986	0.987	0.986	0.986	0.987	**0.987**
Madelon	0.792	0.808	0.783	0.802	0.801	0.763	0.804	0.778
ISOLET	0.990	0.992	**0.992**	0.991	0.990	0.991	0.946	0.991
Gisette	0.986	0.989	0.989	0.986	0.984	0.986	0.972	0.989
Credit-Card	0.980	**0.982**	0.979	0.974	0.982	0.980	0.980	0.979

Table 9: $\mathrm{CR}_{1\%}$, $\mathrm{CR}_{5\%}$ (means over 5 seeds). Best **bold**.

		$\mathbf{CR}_{1\%}$			$\mathbf{CR}_{5\%}$	
Dataset	MinPPI	MOFS-S	NSGA2-u	MinPPI	MOFS-S	NSGA2-u
Spambase	0.714	**0.754**	0.714	0.786	**0.854**	0.821
Madelon	0.674	**0.810**	0.704	0.674	**0.810**	0.702
ISOLET	0.354	**0.382**	0.262	0.654	**0.756**	0.634
Gisette	0.668	**0.782**	0.676	0.682	**0.792**	0.672
Credit-Card	0.710	0.862	**0.880**	0.897	0.883	**0.897**
Crimedata	0.614	**0.808**	0.676	0.646	**0.808**	0.660
Abalone	0.300	0.300	**0.400**	**0.725**	0.725	**0.725**

E Alpha Sensitivity

Harmful counts increase monotonically with α (higher quota $\Rightarrow$ fewer winning coalitions $\Rightarrow$ easier to flip). Decisive features are consistently zero across all α and datasets (consistent with Corollary 1). We recommend $\alpha = 0.95$: it produces meaningful harmful% (1–50%) while $\alpha \leq 0.90$ yields mostly zero signal and $\alpha = 0.99$ is overly selective.

F MinShap Comparison and Madelon Ground Truth

The low Jaccard (0.03–0.20) confirms that binary v_α and continuous $v(S)$ identify fundamentally different feature sets as harmful. MinShap flags 3–8$\times$ more features as negative but with lower precision on the Madelon ground truth; MinPPI is conservative, flagging only features with at least one context where they unambiguously turn a winning coalition into a losing one.

Table 10: Regression results on Crimedata and Abalone ($R^2@k$, means over 5 seeds). Best **bold**.

Dataset	**MinPPI**	**SHAP-k**	**MOFS-S**	**NSGA2-u**	**Banzhaf**	**MinShap**	**ShapG**	**PShap**
				$R^2@10$				
Crimedata	0.660	0.498	0.583	0.577	0.468	0.479	0.400	0.550
Abalone	0.494	0.490	0.490	0.485	0.496	0.474	0.478	0.484
				$R^2@20$				
Crimedata	0.689	0.567	0.674	0.686	0.591	0.543	0.567	0.625
Abalone	0.519	0.513	0.512	0.508	0.520	0.518	0.498	0.507

Table 11: Harmful feature % vs. quota α (single seed 42). "—" = zero harmful.

Dataset	$\alpha = 0.85$	$\alpha = 0.90$	$\alpha = 0.95$	$\alpha = 0.99$
Spambase	—	—	—	10.7%
Madelon	13.0%	9.0%	49.0%	43.0%
ISOLET	3.0%	5.0%	14.0%	24.0%
Gisette	—	—	—	3.0%
Credit-Card	—	—	10.3%	41.4%
Crimedata	—	2.0%	1.0%	—
Abalone	12.5%	—	50.0%	62.5%

G Ablation Full Tables

H Statistical Tests

H.1 Nemenyi Pairwise Significance Matrices

Both Friedman tests are significant, confirming that method differences on AUC rankings are not due to chance. The large Nemenyi CD (≈ 4.8 ranks) means that, with $N = 5$ datasets, pairwise post-hoc separation is difficult—methods within ≈ 5 ranks of each other cannot be declared significantly different. This is a known limitation of the Demsar procedure at small N and is why we emphasize the CD diagram (Figure 3) and the harmful-utility experiment (Section 6.3) as the primary empirical evidence, with Friedman significance as supporting context.

I Empirical Verification of the Estimators

The toy-game verifications supporting Proposition 1 and the Algorithm A/B comparison are collected here.

Table 12: Madelon ground-truth probe recovery (100 features, 97 probes, 5 seeds).

Method	**Precision**	**Recall**	$\mathbf{F}_1$
MinPPI	0.987 ± 0.029	0.214 ± 0.129	0.337 ± 0.150
MinShap	0.979 ± 0.012	0.668 ± 0.050	0.793 ± 0.033
SHAP	0.971 ± 0.031	0.214 ± 0.140	0.336 ± 0.165

Table 13: MinShap vs. MinPPI harmful-set agreement (Jaccard index, means over 5 seeds).

Dataset	**MinPPI** -1	**MinShap** < 0	**Jaccard**
Spambase	1.6	17.4	0.03
Madelon	21.4	66.2	0.20
ISOLET	7.8	53.2	0.06

Table 14: Min vs. mean operator: HV and AUC@10 (means over datasets).

Guidance	**HV**	**AUC@10**
MinPPI	0.033	0.895
Mean-SHAP	0.034	0.913

Table 15: Binary v_α vs. continuous $v(S)$: harmful% distribution (mean over 7 datasets). Continuous flags 46–66%; binary gives α-controlled rates.

$v(\cdot)$ **type**	$\alpha = 0.85$	$\alpha = 0.90$	$\alpha = 0.95$	$\alpha = 0.99$	continuous
Binary	8.0%	15.5%	24.2%	32.3%	—
Continuous	—	—	—	—	50–66%

Table 16: Guidance strength β: HV (mean over 7 datasets). Friedman $\chi^2 = 0.83$, $p = 0.84$ (n.s.).

Metric	$\beta = 0$	$\beta = 1$	$\beta = 2$	$\beta = 5$
HV	0.031	0.036	0.026	0.036

Table 17: Nemenyi post-hoc: "sig" = significantly different at $\alpha = 0.05$. AUC@10: CD = 4.83 ranks ($k = 8$, $N = 5$).

	MOFS-S	**PShap**	**SHAP-k**	**MinShap**	**NSGA2-u**	**Banzhaf**	**MinPPI**	**ShapG**
MOFS-S	—	ns	ns	ns	ns	ns	ns	ns
PShap	ns	—	ns	ns	ns	ns	ns	ns
SHAP-k	ns	ns	—	ns	ns	ns	ns	ns
MinShap	ns	ns	ns	—	ns	ns	ns	ns
NSGA2-u	ns	ns	ns	ns	—	ns	ns	ns
Banzhaf	ns	ns	ns	ns	ns	—	ns	ns
MinPPI	ns	ns	ns	ns	ns	ns	—	ns
ShapG	ns	ns	ns	ns	ns	ns	ns	—

Table 18: Friedman test summary. Kendall's W (effect size) and Nemenyi CD.

Metric	χ^2	p	Kendall's W	CD	#Methods
AUC@10	14.13	0.049	0.35	4.83	8
AUC@20	14.47	0.043	0.36	4.83	8

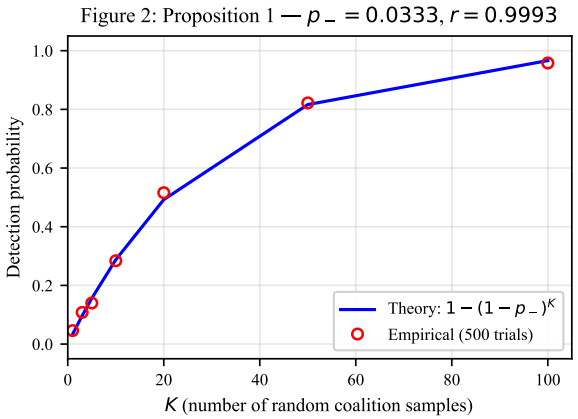


Figure 5: Empirical verification of Proposition 1 on an engineered voting game with $p_-^i = 0.0333$ (500 trials per K). Theory (solid line) and empirical rates (open circles) agree closely ($r = 0.9993$). For sparse harmful contexts ($p_- \ll 1$), Algorithm A requires $K \approx 90$ to reach 95% detection.

Table 19: Quantitative comparison of Algorithm A and Algorithm B (500 trials per setting).

(a) Detection of MinSS $= -1$

K	A detect	B detect	A $\bar{s}$	B $\bar{s}$
1	0.026	1.000	1.00	1.00
2	0.060	1.000	1.98	1.00
3	0.124	1.000	2.87	1.00
5	0.146	1.000	4.69	1.00
10	0.322	1.000	8.43	1.00
20	0.466	1.000	14.91	1.00
50	0.850	1.000	23.77	1.00

(b) Robustness to $\mathcal{N}(0, \sigma^2)$ on W (K=5)

σ	B detect	A detect
0.05	1.000	0.172
0.15	0.878	0.170
0.30	0.652	0.142
0.50	0.508	0.148